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Temporal Pixel Masking for Mitigating OCR-Based Text Recovery from Short-Exposure Screen Photography: A Feasibility Study with a UVC Camera

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[PLACEHOLDER: Funding acknowledgment to be filled.] This manuscript includes a minimal-risk human-subject usability study. The authors' institution does not maintain an ethics review committee for non-medical behavioral experiments of this type. Before each session, the experimenter individually confirmed informed consent and photosensitivity-screening eligibility in person. The study also followed the withdrawal safeguards described in Section V-D, and only de-identified data are analyzed and reported.

ABSTRACT Screen content can be covertly photographed and transcribed by optical character recognition (OCR) or vision-language models (VLMs). We present temporal pixel masking, a display method exploiting differences between human temporal integration and short camera exposures. A base frame is decomposed into rapidly alternating, cryptographically randomized complementary subframes; noise, stripe/glyph overlays, and a partial inversion frame can reduce recoverability. Short rolling-shutter exposures may capture temporally mixed fragments rather than fully integrated frames. We evaluate a 240 Hz software prototype using one USB Video Class (UVC) webcam across nine geometries and 10,575 captures. Across nine archived geometries, best-of-engine OCR character recovery decreased from 94.1% unprotected to 15.1% for the readability-priority profile and 5.0% for the high-suppression profile; exact-match rates decreased from 65.7% to 0.4% and 0%, respectively. The latter meets the descriptive $<$10% recovery target. Video temporal averaging of 150 frames recovered 47.9–71.1% of characters, and the strongest of three tested VLMs reached 77.8% exact match on 0.5 m short-exposure captures, with recovery concentrated on large-font short snippets. Under long exposure, masking could increase recovery over the unprotected baseline, consistent with reduced luminance shifting the sensor into its linear range. These results establish feasibility for mitigating conventional OCR in the tested short-exposure UVC-camera link, but not protection against temporal integration or VLM attackers. In 56 participants, readability-priority masking retained 94.83% typing accuracy but reduced speed by 1.14 WPM; readability, stability, and comfort averaged 3.79, 2.98, and 3.39 out of 5, indicating measurable usability costs.

INDEX TERMS Adversarial perturbation, camera capture, display privacy, optical character recognition, persistence of vision, screen-camera capture, temporal masking, visual eavesdropping

I. INTRODUCTION

SCREENS fill offices, banks, hospitals, and conference rooms, and so do capture devices: smartphones, smart glasses [1] (e.g., Ray-Ban Meta), and miniature cameras. Because a single photograph of a screen suffices to extract text via optical character recognition (OCR) or to identify interface elements through object detection, *visual eavesdropping* is covert, leaving no network traces and requiring no physical contact with the target device. Controlled experiments and survey studies confirm that such attacks are common, succeed at high rates, and frequently go unnoticed [2], [3].

A. EXISTING APPROACHES AND THEIR LIMITATIONS

Current screen-privacy solutions each address a narrow threat subset: spatial, software, watermarking, and temporal approaches cover different viewing geometries or attack stages, but none reports the same combination of on-axis short-exposure capture, static sensitive text, and post-capture machine recognition (§II). That gap motivates measuring machine recovery from physical snapshots, despite the inherent

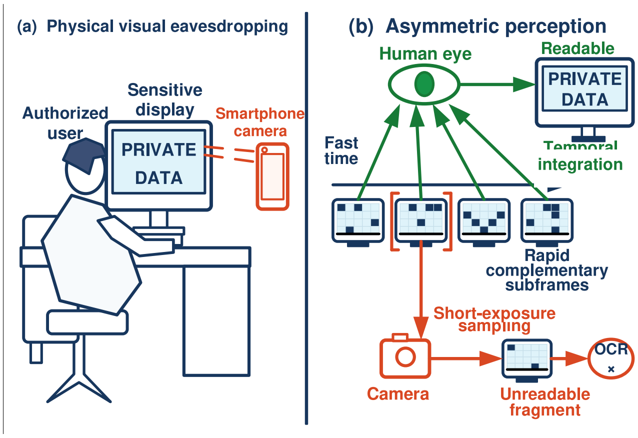

FIGURE 1. Threat scenario and operating principle of temporal pixel masking. (a) A nearby camera can capture sensitive on-screen content without accessing the host system. (b) Human vision integrates complementary subframes, while a short camera exposure can intersect only part of a display cycle and leave OCR with an incomplete fragment. Long exposure or multi-frame reconstruction can accumulate more information.

difficulty that degrading information for the camera also tends to degrade it for the legitimate viewer. We ask whether display temporal structure can widen this gap under short-exposure capture.

B. CORE IDEA

Our approach exploits the gap between human temporal integration and camera exposure windows by decomposing each frame into n randomly complementary subframes that alternate at high speed. When the exposure is shorter than or close to one subframe duration, the camera records only an incomplete pixel subset, whereas the human observer integrates content across subframes (Fig. 1). Long exposure or video reconstruction covering a full cycle can, however, resum the complementary subframes.

This study compares conventional OCR recovery across three display profiles at one USB Video Class (UVC) control setting, with an instrumented pipeline archiving each capture as an image for offline recognition. Across the 8 common-setting geometries, matched mean recovery is 94.5% unprotected, 17.8% for readability-priority, and 5.6% for high-suppression, with the latter meeting its descriptive $< 10\%$ target.

Attack-oriented evaluation expands this result: a temporal-mean attack averaging 150 short-exposure frames (nominally 2.5 s at 60 fps) and vision-language model (VLM) probes recovered considerable large-font text (§V-C), showing where recovery rises beyond the primary conventional-OCR setting.

C. CONTRIBUTIONS

The main contributions of this paper are:

1) **Controlled physical measurement**: Building on prior temporal masking ideas, we implement pixel-level complementary subframes with profile-dependent stripe/glyph overlays and a graphics processing unit (GPU) rendering prototype. The archive contains 10,575 captures from one eMeet SmartCam S600 across 3 distances and 3 angles. On 288 matched common-setting units per profile (12 content items × 8 geometries × 3 repeats), mean recovery is 94.5% unprotected, 17.8% for readability-priority, and 5.6% for high-suppression across geometrically corrected crops processed by three open-source OCR engines.
2) **Cluster-aware evidence and exposure sensitivity**: We report capture-level distributions and content-cluster paired contrasts. Across all 9 archived geometries, unprotected short exposure exceeds readability-priority by 78.0 percentage points (content-cluster resampling interval [75.5, 80.3]) and exceeds high-suppression by 89.1 percentage points ([85.0, 92.4]). Restricting the analysis to the 8 common-setting geometries gives corresponding contrasts of 76.7 points ([74.3, 79.2]) and 88.9 points ([85.5, 91.9]).
3) **Attack and cross-task mapping**: Per-sample strongest digital attack recovery reaches 95.2%. A 150-frame camera mean recovers 71.1% for readability-priority and 47.9% for high-suppression, rising to 79.7% and 53.9% after the underexposed geometry is excluded. VLM probes recover substantial large-font content. Detection and tracking stress tests cover four detectors on 150 real-capture COCO images and 450 MOT still-frame captures, with additional software diagnostics on 8 COCO images and 5,316 MOT frames. Together, these experiments map recovery across text recognition, temporal integration, detection, and tracking.

§II reviews prior work, §III defines the threat model, and §IV presents the design. §V reports OCR evidence, VLM probes, the usability study, detection/tracking stress tests, and simulation diagnostics. §§VI and VII discuss the evidence hierarchy.

II. RELATED WORK

A. SCREEN VISUAL EAVESDROPPING THREATS

Remote screen reading attacks (telescopes, telephoto lenses) [4], deep-learning reconstruction of HDMI electromagnetic leakage [5], covert capture by smart glasses [1], and indirect eavesdropping through reflections and refractions [6] are all documented. In a controlled Ponemon Institute study, 91% of visual-hacking attempts succeeded overall, and the first successful intrusion occurred within 15 minutes in 49% of cases [2]. Eiband et al. [3] surveyed 174 participants on the prevalence of shoulder surfing and the coping strategies users adopt. Tang and Shin [7] proposed Eye-Shield, which re-renders screen content in real time so it stays readable at the user's close viewing distance but appears blurred or pixelated at the larger distances and angles typical of shoulder surfers. Eye-Shield addresses human onlookers; our work instead measures machine recognition applied to short-exposure physical captures.

B. DISPLAY PRIVACY PROTECTION

Physical privacy filters restrict viewing angles but cannot block an on-axis camera. Chen et al. proposed HideScreen [8], a display-side shoulder-surfing defense that suppresses low-frequency visual cues so screen content blends with the background beyond a designed viewing distance, targeting human shoulder surfing in the spatial domain rather than short-exposure camera capture or machine recognition. Zhu et al. [9] proposed LiShield, which modulates smart LED illumination to interfere with rolling-shutter cameras but acts on ambient lighting rather than displayed content. Visual cryptography splits a secret image into shares that combine upon overlay [10], inspiring temporal decomposition ideas despite its original formulation targeting printed media. Digital watermarking [11] can embed tracking identifiers, including screen-shooting-resilient variants [12], but provides post-hoc attribution rather than pre-emptive blocking, and software-based digital rights management (DRM) prevents digital screenshots but not physical photography.

Screen-camera visible light communication (VLC) [13]–[15], which similarly exploits human temporal integration to embed information imperceptible to the eye yet decodable by cameras, is the mirror image of our method: VLC aims to let cameras *read* hidden information, whereas we aim to reduce machine readability of displayed content in constrained short-exposure capture.

Early work by Yerazunis and Carbone [16] examined temporal image masking for display privacy. Wu and Zhai [17] systematized temporal psychovisual modulation, and Gao et al. [18] applied it to information-security displays. Zhang et al. [19] proposed Kaleido, which re-encodes video through multi-frame decomposition. Rainbow [20] instead modulates ambient illumination to corrupt rolling-shutter photographs of physical objects. These schemes use imaging and subjective-quality metrics that are not interchangeable with the recovery measures used here, and the cited studies do not report our specific combination of static text, short UVC exposure, and post-capture recognition. Whereas Kaleido uses chrominance-complementary frames and luminance variation, our implementation uses randomized discrete pixel partitioning, with the exploratory high-suppression profile also abandoning strict completeness (§IV-E). Lim [21] used persistence of vision to generate anti-screen-capture keypads in a browser. Table 1 situates the present study among representative spatial, illumination, and temporal schemes.

The contribution is a system-level UVC measurement spanning explicit-field recovery, temporal integration, and VLM recognition. Detection, tracking, and digital usability proxies provide secondary diagnostics.

C. ADVERSARIAL PERTURBATIONS AND PHYSICAL-WORLD ROBUSTNESS

Digital adversarial perturbations [22]–[24] and physical-world adversarial patches [25]–[27] are surveyed by Wei et al. [28]. Athalye et al. [29] achieved the first cross-viewpoint-robust 3D physical adversarial objects via Expectation over Transformation (EoT), though their approach still requires white-box access to the target model. Physical deployment introduces transformations absent from a digital evaluation, so robustness must be tested across capture conditions rather than inferred from pixel-space performance [25]. Physical adversarial patches exhibit cross-viewpoint robustness but require perturbation patterns pre-generated for specific classifiers [26], [27], with limited transferability to unseen models [30]. The screen-camera link also introduces moiré artifacts, which Niu et al. [31] demonstrated can themselves act as adversarial perturbations. The base mechanism partitions content temporally at the display layer rather than optimizing a model-specific spatial perturbation. Enhanced profiles also add noise and stripe/glyph overlays, whose physical transfer is evaluated empirically rather than assumed.

III. THREAT MODEL

A. PROTECTED ASSETS AND EVALUATED CONTENT

The motivating assets include passwords, financial data, personally identifiable information (PII), source code, and confidential documents. The corpus represents these assets through credentials, digit strings, code, paragraphs, mixed-language snippets containing embedded URL paths, and a document page. Eighteen high-value fields in 9 of 12 items were annotated before scoring; ordinary prose and incidental section numbers were excluded.

B. ATTACKER CAPABILITIES

Attack capabilities do not form a strict monotonic hierarchy but are jointly determined by exposure, geometry, temporal sampling, and post-processing. Results are reported across the following scope:

- **Primary scope**: Instrumented snapshots at the nominal 3.91 ms UVC exposure-control setting, followed by per-frame recognition with Surya, EasyOCR, and Tesseract. The primary result uses raw camera crops. A separate three-engine stress test selects per capture among five predeclared deterministic transforms plus the raw input.
- **Extended attacks**: Long exposure, 150-frame temporal averaging, off-axis capture, phase-aware channel reconstruction, object detection and association, and commercially hosted vision-language models (§V-C).

C. ATTACKER KNOWLEDGE

The attacker *knows* the temporal masking algorithm, the sub-frame count n, and candidate period parameters, but does *not know* the current cycle's mask seed or phase starting point. The cryptographically secure pseudorandom number generator (CSPRNG) provides per-cycle mask randomization to prevent fixed-pattern learning and cross-cycle correlation; it does not provide a cryptographic guarantee against full-cycle averaging attacks. Once an attacker obtains and registers a complete cycle, an unknown seed cannot prevent linear reconstruction. Sparse sampling may disrupt glyph evidence used by conventional OCR, but the present design does not isolate

TABLE 1. **Scope Comparison with Representative Display and Physical-Scene Privacy Schemes. Entries Describe Different Threats and Evaluation Targets; They Are Not Matched-Parameter Performance Claims.**

Scheme	Protected medium	Target threat	Mechanism	Reported evaluation
HideScreen [8]	Mobile display	Human shoulder surfing beyond a designed distance	Spatial suppression of low-frequency cues	Human-viewing distance/readability
LiShield [9]	Physical scene	Rolling-shutter camera capture	Modulated ambient LED illumination	Captured-image interference
Kaleido [19]	Displayed video	Continuous-recording piracy	Perceptually complete chrominance/luminance re-encoding	Peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), and subjective quality
Rainbow [20]	Physical objects	Mobile-camera piracy	Ambient-illumination modulation	Rolling-shutter image degradation
Lim [21]	Browser keypad	Single digital screen capture	Persistence-of-vision partial patterns	Keypad capture demonstration
This work	Static screen text	Short-exposure physical capture followed by machine recognition	Randomized pixel partitioning; high-suppression profile breaks completeness	OCR/VLM recovery and integration boundaries

binarization, segmentation, or recognition as the responsible stage. End-to-end VLMs recovered appreciable text from the same archived fragments (§V-C), so the conventional-OCR observation does not transfer to those models.

D. PROTECTION GOALS

We use tiered design goals under short exposure to interpret the evaluated profiles. High-suppression targets conventional OCR character recovery $< 10\%$ with exact match $= 0\%$, whereas readability-priority targets a relative reduction of at least 80% with exact match $< 1\%$. These descriptive thresholds organize the profile comparison. Object-detection mean average precision at 0.50 intersection over union (mAP50) provides a secondary diagnostic, while ΔE_{00} and the structural similarity index (SSIM) quantify digital reconstruction quality.

IV. SYSTEM DESIGN

A. TEMPORAL SUBFRAME DECOMPOSITION

Given a normalized original frame $\mathbf{I} \in [0, 1]^{H\times W\times C}$, we generate binary masks $\mathbf{M}_1, \ldots, \mathbf{M}_n$ satisfying $\sum_{k=1}^{n} \mathbf{M}_k = \mathbf{1}$, and set $\mathbf{I}_k = \mathbf{I} \odot \mathbf{M}_k$. This yields:

- **Digital-domain completeness**: $\sum_{k=1}^{n} \mathbf{I}_k = \mathbf{I}$;
- **Mutual exclusivity**: each pixel is illuminated in exactly one of the n time slots.

The distinction between "summation" and "temporal averaging" matters: on an ideal linear display with equal slot durations and equal peak luminance, the average radiance over one cycle is $\mathbf{I}/n$, not $\mathbf{I}$, so digital-domain completeness means only that subframes can be re-summed, not that real display luminance has been losslessly preserved.

Pixel assignments are generated from a ChaCha20 [32] CSPRNG, with rejection sampling mapping the byte stream to unbiased slot indices and a separate Fisher–Yates shuffle [33] randomizing subframe playback order. A chi-square slot-count check ($\alpha = 0.01$, df $= n-1 = 3$), used only as an implementation sanity check with at most 5 resampling attempts, is not evidence of cryptographic strength. While randomization reduces fixed patterns and cross-cycle correlation, it cannot prevent linear reconstruction of a registered full cycle. Fig. 2 illustrates the complete pipeline.

B. COMPLEMENTARY ADVERSARIAL NOISE

We overlay adversarial noise $\mathbf{N}_1, \ldots, \mathbf{N}_n$ on the subframes, satisfying:

$$\sum_{k=1}^{n} \mathbf{N}_k = \mathbf{0} \tag{1}$$

Equation (1) holds strictly only in the ideal linear domain prior to summation, clipping, and the display transfer function. Motivated by prior work showing that small text-image perturbations can significantly mislead OCR systems [34], noise directions are generated by fast gradient sign method (FGSM)-style [22] proxy gradients with an amplitude bound of $\varepsilon = 8/255$ and then decomposed complementarily in the temporal domain. Because displays cannot emit negative radiance, the implementation uses a pedestal level and numerical clipping; consequently, after gamma correction, quantization, clipping, and camera imaging, the noise is not guaranteed to cancel physically.

Component role: The random dot-pattern mask is the primary temporal mechanism, while complementary noise is an optional ideal-domain component and stripe/glyph overlays together with partial inversion are nonlinear additions used by the enhanced profiles below. The archived composite-profile definitions preserve correspondence with the captured data, and the physical analysis compares both individual ablations and complete profiles.

C. LUMINANCE COMPENSATION

Because each pixel is illuminated only once per n-slot cycle, a factor-n light-output increase during the active slot would be needed to restore the original cycle-averaged luminance in the linear-light model; the current proof of concept (PoC) uses fixed panel output. CIEDE2000 ΔE_{00} [35] and SSIM [36] quantify digitally integrated and normalized reconstructions across configurations.

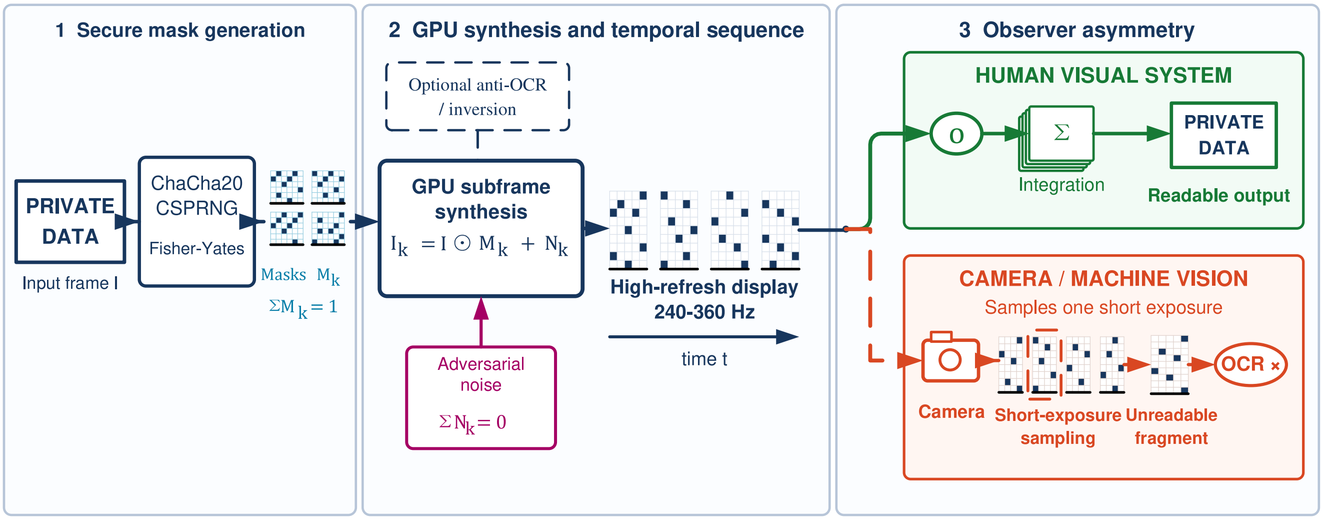

FIGURE 2. Temporal pixel masking system pipeline from source content to protected display. The input frame is decomposed into CSPRNG-assigned complementary pixel masks; optional complementary OCR-oriented noise, profile-dependent stripe/glyph overlays, and an optional partial inversion frame are then inserted before high-refresh playback. A human observer is expected to integrate multiple slots into a readable view, whereas a short-exposure camera may capture only a sparse subframe. Long exposure and video temporal averaging can accumulate multiple slots and are evaluated separately.

D. TIMING AND BANDWIDTH

For a basic cycle containing only n complementary subframes, taking 60 Hz as the minimum cycle rate requires $f_r \geq 60n$; at $n = 4$ this corresponds to 240 Hz. Including one inversion frame changes the requirement to $f_r \geq 60(n+1)$, dropping the actual cycle rate to only 48 Hz at 240 Hz. Thus, while 240 Hz meets the basic four-subframe configuration, it falls short of the inversion-frame configuration's 60 Hz cycle target. Because human flicker perception also depends on luminance, contrast, retinal eccentricity, spatial content, and eye movements [37], [38], nominal cycle rate alone cannot predict comfort.

Whether an observer integrates complementary subframes into a perceptually complete image depends on the temporal contrast sensitivity function (TCSF). Critical flicker fusion (CFF) is not a universal cutoff: Davis et al. [38] found sensitivity near conventional rates for uniform modulation but visibility above 500 Hz when high-frequency spatial edges were present, so the 48 Hz full-cycle rate of the inversion configuration may produce visible flicker for some observers, and the 60 Hz basic configuration cannot be assumed flicker-free. The user study in §V-D evaluates readability and immediate comfort under these configurations.

E. EXPLORATORY HIGH-SUPPRESSION PROFILE

Because the basic linear temporal mask leaks under full-cycle temporal averaging (see §V-F2), we introduce nonlinear enhanced profiles. Unless otherwise stated, profiles in this subsection use $n = 4$ base temporal masking with complementary adversarial noise enabled. Table 2 summarizes all profile configurations; the "Mask + noise" row in Table 6 is the base condition without stripe/glyph anti-OCR overlay.

- **Strong (anti-OCR) profile**: base mask+noise with mask cell size of 1 pixel, stripe width of 10 pixels, stripe amplitude $\alpha_s = 0.10$, and glyph amplitude $\alpha_g = 0.12$, without an inversion frame.
- **Readability-priority profile**: the Strong profile plus an $\alpha = 0.2$ inversion frame, designated in this paper as the usability-facing composite configuration. Archived result tables and capture metadata retain the legacy label "deployed."
- **High-suppression profile**: an exploratory security-oriented stress profile using base mask+noise with mask cell size increased to 2 pixels, stripe width of 6 pixels, stripe and glyph amplitudes of 0.42 and 0.55 respectively, plus an $\alpha = 0.2$ inversion frame.

The nonlinear profiles depart from strict completeness ($\sum \mathbf{I}_k \neq \mathbf{I}$), producing visible stripes and grain. This trade-off is the opposite of Kaleido-class temporal re-encoding: the latter relies on perceptually complete mixing within a cycle to preserve viewing experience, so full-cycle integration reconstructs an approximation of the original. In the static-content capture threat addressed here, completeness is the entry point for integration attacks. The high-suppression profile therefore trades visible artifacts for lower measured recovery under full-cycle integration.

The paper-facing name "high-suppression" reflects the profile's relative recovery in Table 4. The archived identifier remains `capture_hardened` (with legacy tag `vlm`) for reproducibility; §V-B reports residual leakage and behavior under the preprocessing oracle.

F. PARTIAL INVERSION FRAME

A long-exposure attack occurs when the exposure window covers a full display cycle ($n = 4$ at 240 Hz yields a basic

TABLE 2. Profile Composition Summary ($n = 4$ for All Profiles)

Profile	Mask	Noise	Cell (px)	Stripe width (px)	Stripe α_s	Glyph α_g	Inv. α
Mask only	✓	–	1	–	–	–	–
Mask + noise	✓	✓	1	–	–	–	–
Strong	✓	✓	1	10	0.10	0.12	–
Readability-priority	✓	✓	1	10	0.10	0.12	0.2
High-suppression	✓	✓	2	6	0.42	0.55	0.2

TABLE 3. Inversion Frame Strength α Ablation on the Fixed Strong Overlay ($\alpha_s = 0.10$, $\alpha_g = 0.12$): Real-Capture Best-of-Engine Character Recovery (Mean and 95% Capture-Resampling Interval, Five Account/Code/CET6 Items Pooled Across 9 Geometric Conditions; $N = 135$ per Setting for Long Exposure, $N = 45$ for Video Temporal Averaging). This Five-Item Subset Differs from the 12-Item Readability-Priority Rows in Table 6.

α	Long-exp. char (%)	Video avg. char (%)
0.0	68.6 [63.1, 73.9]	72.7 [63.3, 81.1]
0.2	67.0 [61.3, 72.7]	69.8 [59.9, 78.7]
0.3	72.0 [66.2, 76.9]	64.3 [54.0, 73.6]
0.5	61.7 [55.6, 67.7]	66.4 [56.5, 75.2]
1.0	18.1 [14.0, 22.0]	28.4 [20.5, 36.9]

cycle of approximately 16.7 ms; with an inversion frame the $n+1$ slot cycle is approximately 20.8 ms), where the archived long-exposure labels of 31.25 and 125 ms are nominal values derived from UVC controls rather than photometric measurements. To counter this threat in the normalized domain $\mathbf{I} \in [0, 1]$, the implementation inserts a **partial inversion frame** $\alpha(\mathbf{1} - \mathbf{I})$ after each n-subframe cycle. The equivalent 8-bit expression is $\alpha(255 - \mathbf{I}_8)$. Higher α changes both the integrated signal and the digital reconstruction's luminance and contrast.

A real-capture inversion-strength ablation (Table 3; pipeline per §V-B) found 68.6% recovery at $\alpha = 0$ and 67.0% at $\alpha = 0.2$. Point estimates across $\alpha \in [0, 0.5]$ ranged from 61.7% to 72.0%; the capture-resampling intervals are descriptive and interval overlap is not an equivalence test. The ablation holds the Strong overlay fixed and uses a five-item account/code/CET6 subset. Its $\alpha = 0.2$ value ($N = 135$) is not a repeated estimate of the readability-priority row. Near-full inversion ($\alpha = 1.0$) reduced observed recovery to 18.1% while severely degrading the digital integrated view. Under the 150-frame temporal mean, settings at $\alpha \leq 0.5$ retained 64–73% recovery. The readability-priority choice of $\alpha = 0.2$ is a modeled design decision, not a validated optimum.

Each display cycle thus contains $n + 1$ time slots; the resulting 48 Hz cycle rate at $n = 4$ and 240 Hz, and its flicker implications, are discussed in the timing analysis above.

G. IMPLEMENTATION SCOPE

The current PoC operates at the application layer, using Python and GPU rendering to generate display sequences, an architecture that provides direct control over subframe composition, playback order, and profile parameters during evaluation.

V. EXPERIMENTAL EVALUATION

A. EXPERIMENTAL SETUP

Hardware platform: Most experiments used a desktop workstation with an Intel Core i7-8700K (3.70 GHz), 32 GB RAM, and an NVIDIA TITAN Xp (12 GB VRAM). The display was an AOC 24G51Z driven at a nominal 240 Hz. Each nominal 240-Hz slot lasts 4.17 ms, while the basic and inversion-augmented cycles are 60 and 48 Hz.

Capture device: All physical captures used one eMeet SmartCam S600 USB webcam with its standard UVC driver. Camera frames underwent perspective correction and content cropping before recognition. Per-capture metadata store exposure labels derived from DirectShow/UVC $\log_2$ controls: -11 and -8 map nominally to 0.49 and 3.91 ms, while -5 and -3 map to 31.25 and 125 ms. Eight geometries used the common $-8/-5$ settings; d0.5/15° used $-11/-3$. These values are control-derived labels. The display-brightness setting remained fixed, and video was captured nominally at 60 fps. The primary OCR inputs are geometrically corrected content crops.

The camera was placed at 0.5 m, 1 m, and 1.5 m from the display, rotated to 0°, 15°, and 30° at each distance, providing nine distance–angle configurations for the physical evaluation. The manufacturer specifies an 8-megapixel 4K sensor and 1080p/60 fps output [39].

Test content: Each position used 12 fixed items: two account strings, two code snippets, two digit strings, one English sentence, two mixed-language snippets, two English paragraphs, and one CET6 (College English Test Band 6) document. The CET6 page and two mixed-language items include Chinese characters. Three repeats give 36 captures per profile–attack–position cell and 324 across 9 positions before disclosed repeats. Detection and tracking use separate subsets as cross-task stress tests.

Evaluation metrics: OCR character recovery is defined as $\max(0, 1 - \mathrm{CER})$, where CER is character error rate. Exact match uses case-insensitive full-text comparison after whitespace removal. Explicit-field recovery uses 18 fields fixed before scoring across 9 field-bearing content items. A field counts as recovered only when its complete Unicode-normalized form appears in at least one engine output for that capture; case, whitespace, and punctuation are ignored. We report field-level micro recovery and sample-level macro recovery separately. Leak rate is the proportion of samples with character recovery $\geq 20\%$ and is only a descriptive threshold. On the matched common-setting units, readability-priority P50, P75, P90, P95, and P99 character recovery are

10.1%, 22.0%, 45.6%, 63.6%, and 95.0%. The corresponding high-suppression values are 3.4%, 9.2%, 15.2%, 19.1%, and 22.6%.

OCR aggregation and uncertainty: Character and whole-text metrics are maximized independently across the three engines for each capture. Explicit fields use the union of fields recovered by the three engines. Duplicate captures with the same profile, content, position, and repeat are averaged before matching. The primary comparison uses keys present in all three profiles. Paired contrasts resample the 12 content items as clusters with 2,000 resamples and seed 20260612. These intervals quantify sensitivity across the archived content clusters while holding geometry fixed. Table 6 gives capture-resampling summaries for the unbalanced archive. Across all geometries, unprotected recovery exceeds readability-priority by 78.0 percentage points ([75.5, 80.3]) and high-suppression by 89.1 points ([85.0, 92.4]). Mask-only exceeds readability-priority by 3.0 points ([-0.5, 6.5]).

Common-setting analysis: The d0.5/15° position used a different nominal UVC setting and produced visibly underexposed protected frames. The remaining 8 geometries define the primary fixed-setting result (Table 4). After duplicate averaging and three-profile matching, each profile has $N = 288$ units. Mean recovery is 94.5% unprotected, 17.8% readability-priority, and 5.6% high-suppression. The paired contrasts are 76.7 percentage points ([74.3, 79.2]) and 88.9 points ([85.5, 91.9]). The unbalanced captured-image mean for readability-priority is 16.7% across 408 images and is reported only as a sensitivity summary. Excluding the noncomparable geometry raises readability-priority 150-frame video-mean recovery from 71.1% to 79.7% and high-suppression from 47.9% to 53.9%.

Additional robustness checks: A leave-one-round-out analysis on the five repeated content items gives readability-priority short-exposure recovery of 16.6% for round 1 and 15.8% for round 2, leaving the profile-level pattern unchanged. Section V-E reports detection and tracking as cross-task stress tests.

Fixed preprocessing stress test: We evaluated Tesseract, EasyOCR, and Surya on all 984 selected primary-pool images. The fixed input grid comprised raw, gamma 0.5, luminance contrast-limited adaptive histogram equalization (CLAHE; 1st–99th percentile stretch, clip limit 4.0, 8×8 tiles), unsharp masking ($\sigma = 1.0$, weights 1.7/-0.7), Gaussian adaptive thresholding (block 31, constant 5), and $2\times$ bicubic upscaling. Parameters were fixed before scoring. For each metric and capture, the attacker retained the best engine–input pair; duplicate captures were then averaged before the same three-profile matching used by the primary analysis. The complete matrix contains 17,712 cells (984 images $\times$ 6 input forms $\times$ 3 engines) with no OCR errors. Nonraw Tesseract cells used Tesseract 5.4.1 through `pytesseract 0.3.13`; EasyOCR 1.7.2 and Surya 0.14.7 transformed cells used CUDA on the workstation above. The resulting oracle measures a reproducible fixed-grid preprocessing attack.

Digital reconstruction proxies: ΔE_{00} is the mean CIEDE2000 color difference between the original image and the full-cycle digitally reconstructed image; SSIM [36] is their mean structural similarity. Both metrics support within-paper comparison of digitally reconstructed profiles.

B. REAL-CAPTURE OCR EXPERIMENT

The physical archive comprises 10,575 images (1,175 per position), including ablations, parameter sweeps, and main attack conditions. Readability-priority includes a second capture round for 5 of 12 items, giving 459/153 all-position short/video images rather than 324/108. Duplicate keys are averaged before matched contrasts. Surya [40], EasyOCR [41], and Tesseract [42] produced 31,725 engine-level records. The Surya rerun used `surya-ocr 0.14.7`; the primary pipeline also used Tesseract 5.4.1 (`pytesseract 0.3.13`) and EasyOCR 1.7.2. Each geometry uses an archived four-point homography and output canvas, followed by the recorded content crop. The primary OCR pipeline adds no adaptive enhancement. Profiles were collected in the order original, mask-only, mask+noise, Strong, readability-priority, and high-suppression. The main comparison uses 8 common-setting geometries, while Table 6 retains all 9 as sensitivity evidence. Fig. 3 shows digital reconstructions and matched captures at 1.0 m/0°.

Key findings:

1) **Primary fixed-setting evidence**: Table 4 is the main short-exposure estimate. On the same 288 matched units per profile (12 content items $\times$ 8 geometries $\times$ 3 repeats), mean recovery is 94.5% unprotected, 17.8% for readability-priority, and 5.6% for high-suppression. The paired content-cluster contrasts are 76.7 percentage points ([74.3, 79.2]) and 88.9 points ([85.5, 91.9]). The full 9-geometry captured-image pool gives 94.1%, 15.1%, and 5.0% across the archived settings and repeats.
2) **Fixed preprocessing defines a stronger OCR attack**: On the same 288 matched units per profile, the raw three-engine oracle yields 94.5% unprotected, 17.8% readability-priority, and 5.6% high-suppression recovery. Allowing the five fixed transforms raises these values to 95.9%, 40.2%, and 13.7%, respectively (Table 8). The paired unprotected-minus-readability contrast remains 55.7 percentage points ([50.3, 60.5]), while the unprotected-minus-high-suppression contrast remains 82.2 points ([78.5, 85.2]). The complete 17,712-cell matrix has no OCR errors.
3) **Profiles show distinct component and target behavior**: The mask-only condition reaches 19.2% in the pooled archive, while mask+noise reaches 25.6% and Strong reaches 20.7%, revealing nonmonotonic component effects. In the matched pool, readability-priority retains 17.7% field-micro exact recovery (Table 5) and 95.0% P99 character recovery. High-suppression reaches a 5.6% matched mean at the common setting, meeting both its descriptive $< 10\%$ character target and the exact-match $= 0\%$ target: no short-exposure cap-

TABLE 4. Primary Matched Short-Exposure OCR Comparison at the Common Nominal 3.91-ms UVC Setting. Duplicate Captures Are Averaged Before Matching; N = 288 per Profile Comprises 12 Content Items × 8 Geometries × 3 Repeats. The Noncomparable d0.5/15° Run Is Excluded.

Profile	N	Mean char recovery	Unprotected − profile	Interpretation
Original (unprotected)	288	94.5%	–	Common-setting baseline
Readability-priority	288	17.8%	76.7 pp [74.3, 79.2]	Conventional-OCR reduction, with upper-tail failures
High-suppression	288	5.6%	88.9 pp [85.5, 91.9]	Exploratory stress profile; meets descriptive < 10% target

TABLE 5. Explicit Sensitive-Field Exact Recovery in the Primary Matched Short-Exposure Pool. The 18 Fields Were Fixed Before Scoring; 216 of 288 Units per Profile Contain at Least One Field.

Profile	Field micro (432 opp.)	Sample macro
Original (unprotected)	91.4%	86.8%
Readability-priority	17.7%	17.5%
High-suppression	1.2%	1.1%

ture produced an exact match (Table 6). For readability-priority, the raw common-setting reduction is $(94.5 - 17.8)/94.5 = 81.2\%$ with 0.4% exact match, meeting the $\geq 80\%$ relative reduction and $< 1\%$ exact-match targets under the primary analysis. The preprocessing oracle reduces the gap to $(95.9 - 40.2)/95.9 = 58.1\%$, falling short of the same threshold.

4) **Integration attacks recover substantial text** (Fig. 4): The readability-priority profile reaches 60.9% char / 36.4% exact under nominal long exposure and 71.1% char / 42.5% exact under the 150-frame mean. Excluding d0.5/15° raises video recovery to 79.7%. Weak inversion ($\alpha = 0.2$) produces little integration reduction, and mask-only and Strong are similar under the 150-frame mean (79.0% versus 78.6%). High-suppression shows lower integration recovery, although substantial text remains recoverable.
5) **Long exposure produces a position-dependent reversal**: The unprotected long-exposure baseline is 47.3% char, below the readability-priority profile's 60.9%. At the sole 125 ms position (0.5 m/15°), unprotected recovery is 1.1% versus 46.3% readability-priority, with visible clipping in the full-brightness image. At 31.25 ms, the expected direction holds at 0.5 m, whereas five of six 1.0–1.5 m positions show higher readability-priority recovery. Residual stripe/glyph edges interacting with geometry-specific saturation or contrast compression provide a candidate explanation. Short-exposure recovery remains lower for readability-priority at every position.
6) **High-suppression reduces integration leakage**: This exploratory stress profile yields 9.3% char [7.9, 10.9] / 0% exact for nominal long exposure and 47.9% char [41.8, 54.0] / 5.6% exact [1.9, 10.2] under the 150-frame mean. Brackets are capture-resampling summaries. The pooled mean includes considerable inter-position variation and a 0% underexposed session. Excluding that session raises video recovery to 53.9% [48.0, 59.9] and exact match to 6.3%. The user study evaluates readability-priority rather than this visibly stronger profile.
7) **Geometric trends**: Across 9 conditions (0.5–1.5 m × 0–30°), protected short-exposure recovery remains below unprotected recovery. Geometry-specific values are reported descriptively because the archived design is unbalanced.

Per-engine results: Table 7 breaks down pooled raw-input short-exposure results by engine. The engines differ widely in unprotected recovery (Surya 37.1%, EasyOCR 79.5%, Tesseract 84.3%). Under Strong, recovery ranges from 2.1–16.3%; under readability-priority, 3.2–11.7%; under high-suppression, 1.8–2.0%. Table 8 and Fig. 5 use the matched common-setting pool and show that every engine benefits from at least one fixed transform. For readability-priority, recovery rises from 3.2% to 10.8% for Tesseract, 14.2% to 33.9% for EasyOCR, and 4.1% to 20.2% for Surya. For high-suppression, the corresponding changes are 2.0% to 8.2%, 2.3% to 6.5%, and 2.0% to 6.0%.

C. REAL-CAPTURE VLM PROBES

To test temporal masking against stronger recognizers, we applied three commercially hosted VLM endpoints exposed through the SiliconFlow application programming interface (API): Qwen3-VL-32B-Instruct [43], Kimi-K2.6 [44], and GLM-4.5V [45], thereby showing how recovery varies with model capability on the same photographs.

The probes cover 0° on-axis at three distances. The 60 fps video was split into 12 equal segments and aggregated four ways: (1) *single best*: the single frame within each segment with the highest grayscale contrast; (2) *150-frame mean*: the pixel-wise average of all 150 frames in the segment; (3) *window-mean best*: a sliding 30-frame mean, keeping the window with the highest mean luminance; and (4) *max projection*: the per-pixel temporal maximum across all frames in the segment. The 0.5 m run includes 156 inputs: 36 unprotected short-exposure baselines, 36 high-suppression short- and long-exposure inputs each, and these four aggregate views (12 segments each). The 1.0 m and 1.5 m runs each include 120 high-suppression inputs: 36 short- and long-exposure inputs each plus four video aggregate types (12 segments each). VLM inputs and the OCR best-of-engine (BoE) reference in Table 9 use the same file batch with nominal 3.91/31.25 ms labels. All VLM calls use raw camera JPEGs.

Of 1,188 model requests, 1,133 returned parseable results:

TABLE 6. Single-UVC Real-Capture OCR Sensitivity Pool (Best-of-Engine; N = Captures per Row; All 9 Geometries). Brackets Are 95% Capture-Resampling Summaries, Not Population Confidence Intervals. Video Rows Average 150 Frames. The Primary Matched Estimate Is Table 4.

Profile	Attack mode	N	Char recovery [95% summary]	Exact match	Leak rate
Original (unprotected)	Short exposure	324	94.1 [93.1, 95.0]%	65.7%	99.4%
	Video (150-frame mean)	108	79.9 [73.5, 86.0]%	61.1%	86.1%
	Long exposure	324	47.3 [42.8, 51.9]%	18.8%	58.6%
Mask only	Short exposure	324	19.2 [17.0, 21.4]%	0.3%	30.6%
	Video (150-frame mean)	108	79.0 [73.0, 84.9]%	48.1%	88.0%
	Long exposure	324	67.2 [63.0, 71.5]%	35.5%	75.0%
Mask + noise	Short exposure	324	25.6 [22.5, 28.8]%	2.8%	36.1%
	Video (150-frame mean)	108	79.2 [73.3, 85.0]%	49.1%	88.9%
	Long exposure	324	68.0 [63.7, 72.4]%	39.8%	76.2%
Strong (anti-OCR)	Short exposure	324	20.7 [18.2, 23.3]%	0.6%	32.4%
	Video (150-frame mean)	108	78.6 [72.5, 84.5]%	50.9%	88.9%
	Long exposure	324	61.6 [57.1, 66.1]%	39.8%	71.3%
Readability-priority	Short exposure	459	15.1 [13.3, 17.1]%	0.4%	20.5%
	Video (150-frame mean)	153	71.1 [65.7, 76.2]%	42.5%	85.0%
	Long exposure	324	60.9 [56.6, 65.6]%	36.4%	69.1%
High-suppression	Short exposure	324	5.0 [4.3, 5.8]%	0.0%	3.7%
	Video (150-frame mean)	108	47.9 [41.8, 54.0]%	5.6%	76.9%
	Long exposure	324	9.3 [7.9, 10.9]%	0.0%	14.2%

TABLE 7. Real-Capture Short-Exposure OCR: Per-Engine Results (Pooled Across 9 Geometric Conditions)

Profile	Engine	Char (%)	Exact (%)	N
Original	Surya	37.1	22.8	324
	EasyOCR	79.5	27.8	324
	Tesseract	84.3	47.8	324
Mask only	Surya	4.2	0.3	324
	EasyOCR	15.6	0.0	324
	Tesseract	3.2	0.0	324
Mask + noise	Surya	11.0	2.8	324
	EasyOCR	18.1	0.0	324
	Tesseract	4.8	0.0	324
Strong	Surya	6.1	0.6	324
	EasyOCR	16.3	0.0	324
	Tesseract	2.1	0.0	324
Readability-priority	Surya	3.2	0.4	459
	EasyOCR	11.7	0.0	459
	Tesseract	3.3	0.0	459
High-suppression	Surya	1.8	0.0	324
	EasyOCR	2.0	0.0	324
	Tesseract	1.9	0.0	324

TABLE 8. Matched Common-Setting Character Recovery Under the Fixed Preprocessing Grid. Each Cell Is Raw Input → Per-Capture Grid Oracle (%); N = 288 per Profile After Duplicate Averaging and Matching.

Engine	Original	Readability	High-supp.
Tesseract	84.9 → 91.8	3.2 → 10.8	2.0 → 8.2
EasyOCR	85.1 → 88.4	14.2 → 33.9	2.3 → 6.5
Surya	30.6 → 49.3	4.1 → 20.2	2.0 → 6.0
Best of three	94.5 → 95.9	17.8 → 40.2	5.6 → 13.7

34 failures at 0.5 m, 21 at 1.0 m, and none at 1.5 m. Each table cell reports effective N and is conditional on a parseable response. GLM failures concentrate on mixed, digit, and credential items, so missingness cannot be treated as random. We computed cell-wise lower and upper bounds by assigning every failed call 0 or 100% recovery. For the GLM 0.5 m short cell, character recovery is bounded by 61.0–74.9% and exact recovery by 44.4–58.3%. Conditional cells are not used to rank models; complete bounds for all cells are archived with the analysis.

We use the failure-free 1.5 m run as primary cross-model evidence; the 1.0 m run enters the short-exposure distance sweep as supplementary data. At 1.0 m, Qwen and Kimi long-exposure exact match are 84.4% and 74.3%, and the successful calls from the three models yield 70.0–83.3% exact match under the 150-frame video mean. GLM long-exposure exact match is only 3.3% at 1.0 m versus 58.3% at 1.5 m, but content-dependent failures preclude a cross-distance consistency claim. For compactness, Table 9 reports only the 150-frame video mean at 0.5 m; the 1.5 m run reports all four aggregate views to show view-selection sensitivity.

All calls used SiliconFlow's OpenAI-compatible API. The recorded provider identifiers are `Qwen/Qwen3-VL-32B-Instruct`, `Pro/moonshotai/Kimi-K2.6`, and `zai-org/GLM-4.5V`. Qwen3-VL and GLM-4.5V have public technical reports [43], [45]. For Kimi-K2.6, the public model card [44], the provider endpoint identifier, and archived request metadata form the reproducibility record. Decoding temperature is 0 and maximum output is 4,096 tokens. The shared prompt requests only visible text and prohibits guessing; ground truth is not provided. Full prompts, parameters, and per-call outputs are archived. Metrics follow §V-A and are pooled at the capture level.

0.5 m session selection transparency: The archive preserves an alternative session (`real_capture_vlm_d0.5_a0_rerun.json`, timestamp 012715) at the same position whose high-suppression short-exposure inputs are nearly

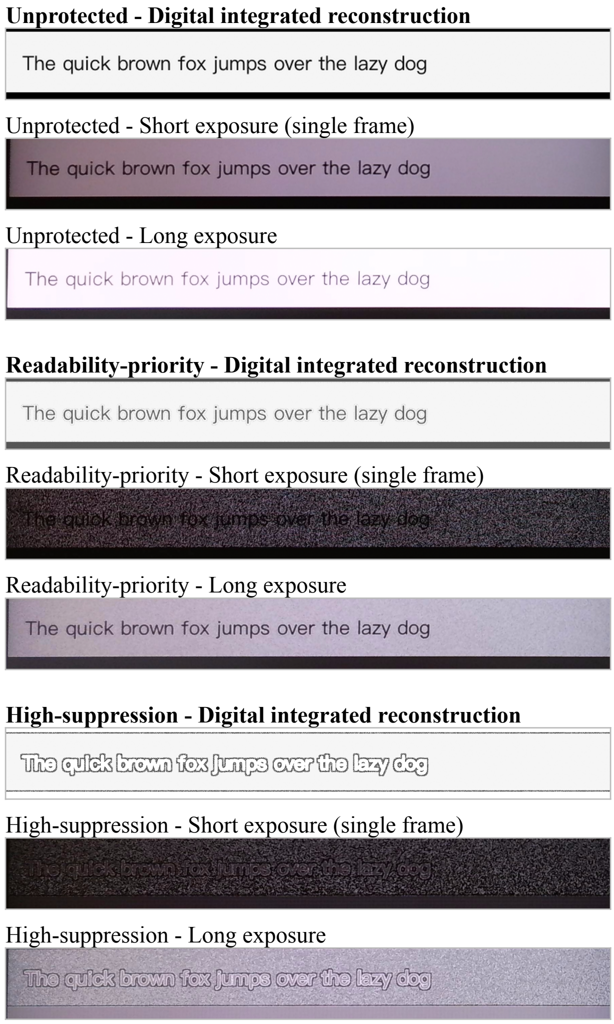

FIGURE 3. Real-capture montage at 1.0 m and 0°. Within each vertically stacked profile group (unprotected, readability-priority, and high-suppression), panels show, from top to bottom, the digital integrated reconstruction, 3.91 ms short-exposure capture, and 31.25 ms long-exposure capture. Camera panels are identically cropped from the archived raw camera JPEGs with no geometric correction or brightness gain, so they differ from the geometrically corrected crops used for OCR evaluation. Digital panels use the same 1920 × 1080 black playback canvas and brightness-aligned full-cycle integration for each profile (for the unprotected profile this integration equals the displayed source frame); the readability-priority and high-suppression reconstructions include the recorded $\alpha = 0.2$ inversion slot. These digital views are not direct evidence of human readability or visual comfort.

all-black, yielding empty transcriptions and 0% char/exact across all three models. Table 9 uses the 141107 (3.91 ms) session because it shares the OCR reference batch and its inputs are not black. These are the only two 0.5 m on-axis VLM sessions, so Qwen's observed short-exposure exact-match outcome ranges from 0/36 captures in the underexposed session to 28/36 captures in the tabled session, not a stable single-session estimate. The adequate-exposure criterion was applied after inspecting two sessions and is therefore a post-hoc selection rule; preserving both sessions and reporting the range makes the exposure sensitivity explicit. The tabled session is the conservative attacker-favorable case, while the alternative is retained as an underexposure diagnostic.

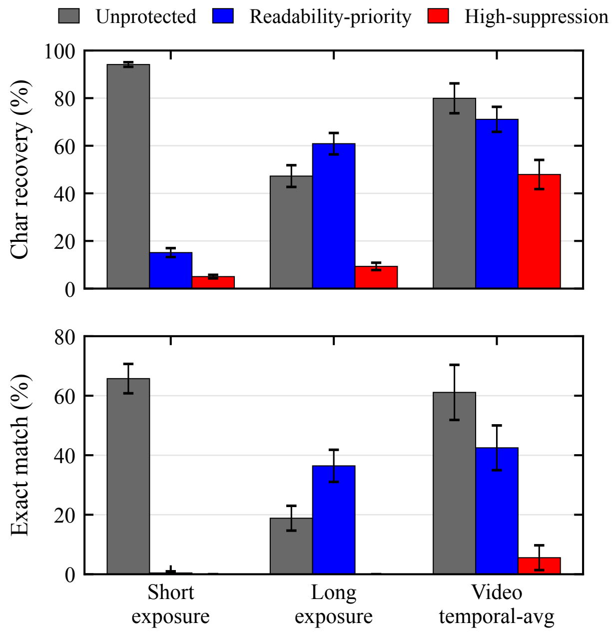

FIGURE 4. Best-of-engine OCR character recovery and exact match for three representative real-capture profiles (unprotected, readability-priority, and high-suppression) under short exposure, long exposure, and a 150-frame video temporal mean, pooled across the 9 tested distance–angle configurations. Lower bars indicate less recovered text. The long-exposure and video bars expose the integration boundary: the readability-priority profile retains substantial recoverable content even when its short-exposure bar is low. Readability-priority sample counts include the disclosed five-item repeat and therefore differ from the other profiles.

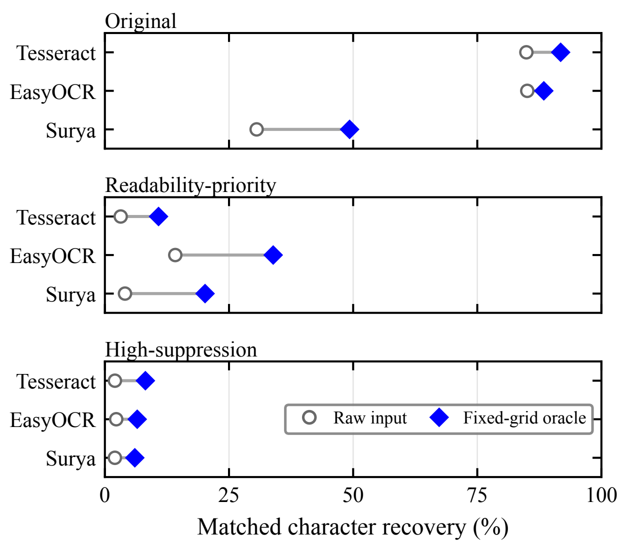

FIGURE 5. Raw-input versus fixed-grid-oracle character recovery for each OCR engine on the matched common-setting pool (N = 288 per profile). The oracle selects the best of raw plus five predeclared transforms separately for each capture and metric. Lines show within-engine increases, not uncertainty intervals. Exact values appear in Table 8.

TABLE 9. Real-Capture VLM Probes (0° On-Axis; All Rows Except the Original Short Baseline Are High-Suppression). Each VLM Cell Reports Exact Count / Char / N (Recoveries / % / Effective Calls); Cells with Failures Are Conditional on Parseable Responses. OCR BoE Shows the Same-Batch Three-Engine Best-of-Engine Reference (Exact / Char, %). The Temporal-Mean Rows Average 150 Short-Exposure Frames; Other Aggregate Views Are Defined in the Text. The 0.5 m High-Suppression Short Row Uses the Adequate-Exposure Attacker-Favorable 141107 Session; the Only Alternative 0.5 m Session Was Underexposed and Yielded 0/36 Qwen Exact Matches.

Distance	Condition	OCR BoE	Qwen3-VL-32B	Kimi-K2.6	GLM-4.5V
Short-exposure distance sweep					
0.5 m	original short	58.3 / 92.4	30 / 94.5 / 36	30 / 95.6 / 36	19 / 92.0 / 26
0.5 m	short	0.0 / 8.4	28 / 91.5 / 36	17 / 84.7 / 36	16 / 70.8 / 31
1.0 m	short	0.0 / 3.6	20 / 79.1 / 36	14 / 68.6 / 36	9 / 47.0 / 33
1.5 m	short	0.0 / 6.5	17 / 68.9 / 36	12 / 61.5 / 36	13 / 70.0 / 36
Key attack views					
0.5 m	long	0.0 / 2.1	28 / 85.7 / 34	1 / 7.9 / 36	0 / 4.9 / 30
0.5 m	Video (150-frame mean)	8.3 / 50.7	10 / 94.0 / 12	10 / 95.5 / 12	7 / 91.8 / 8
1.5 m	long	0.0 / 13.2	33 / 89.3 / 36	32 / 89.9 / 36	21 / 62.4 / 36
1.5 m	Video (single best)	0.0 / 4.3	1 / 24.8 / 12	0 / 12.7 / 12	0 / 29.8 / 12
1.5 m	Video (150-frame mean)	0.0 / 48.4	8 / 90.3 / 12	8 / 90.3 / 12	7 / 88.5 / 12
1.5 m	Video (window-mean best)	0.0 / 8.6	7 / 89.0 / 12	5 / 84.2 / 12	8 / 87.2 / 12
1.5 m	Video (max projection)	0.0 / 19.0	8 / 90.1 / 12	7 / 89.2 / 12	7 / 88.7 / 12

TABLE 10. 0.5 m High-Suppression Short-Exposure VLM Character Recovery by Content Type. Baseline Column Shows Same-Session Unprotected Original Short Results for Qwen (Kimi/GLM Similar); GLM Had Occasional Call Failures, Parenthesized N Shows Effective Count. At 1.5 m, CET6 Dense Pages Yield 0% Char Across All Three Models.

Content type (N)	Baseline (%)	Qwen (%)	Kimi (%)	GLM (%)
CET6 document (3)	64.4	18.5	5.6	0.4
Credentials (6)	97.6	97.6	88.3	77.1 (5)
Code snippets (6)	89.8	94.9	95.8	88.9
Digit strings (6)	100.0	100.0	98.6	50.0 (4)
Mixed content (6)	100.0	100.0	92.5	98.2 (4)
Paragraphs (6)	100.0	99.8	81.7	66.5
English sentences (3)	95.3	95.3	96.9	94.6

Table 9 presents the pooled VLM probe results, where short-exposure rows show distance variation while long-exposure and video aggregate rows expose integration attack boundaries; Table 10 breaks down 0.5 m high-suppression short-exposure character recovery by content type.

Key findings:

1) **VLMs recover substantially more text than conventional OCR**: On the same 0.5 m photographs, best-of-engine OCR yields 8.4% char / 0% exact on high-suppression short exposure, while Qwen3-VL achieves 91.5% char and exactly recovers 28/36 captures. At 1.5 m, the three models retain 33.3–47.2% exact / 61.5–70.0% char. The gap between conventional OCR and modern VLMs is large.
2) **VLM recovery concentrates on large-font short snippets**: At 0.5 m, CET6 document pages yield 0.4–18.5% VLM character recovery, whereas credentials, code, and digit strings yield 50–100% (Table 10). At 1.5 m, CET6 pages drop to 0% across the three models, while digit strings and short sentences remain recoverable. The pattern is consistent with font scale and language priors [46], [47].
3) **Attack capability varies across models and conditions**: In the failure-free 1.5 m session, nominal long-exposure exact recovery ranges from 58.3% to 91.7%, with 62.4–89.9% character recovery. Endpoint, distance, capture view, and content jointly shape this spread.
4) **Sustained video aggregation strongly increases VLM recovery**: At 0.5 m, missingness bounds for the 150-frame mean span 58.3–91.7% exact and 61.2–95.5% character recovery across models. In the failure-free 1.5 m session, the 150-frame mean yields 58.3–66.7% exact and 88.5–90.3% character recovery, while single-best-frame exact recovery is 0–8.3%. The accompanying 12.7–29.8% single-best character values rest on low-information inputs, where fluent language-prior completions can inflate character recovery even though the prompt prohibits guessing; the exact-match counts are the more conservative reading.

Although high-suppression strongly reduces dense small-text recovery under conventional OCR, VLMs still recover many large-font credentials, code snippets, and digit strings, and content-adaptive, VLM-aware temporal profiles warrant investigation.

D. USER EXPERIENCE STUDY

To evaluate the impact of the protected display on human readability and visual comfort, we designed and conducted a web-based user study.

Equipment and environment protocol: The same 240 Hz display as in §V-A. The formal study version sets a hard measured refresh-rate threshold of ≥200 Hz to ensure the fixed $n = 4$ basic subframe cycle rate $f_r/n \geq 50$ Hz; below this threshold participation is blocked rather than silently reducing n. The 144–199 Hz adaptive path is available only for an explicitly labeled demonstration mode, and its sessions are excluded from analysis by default. Before each participant begins, the experimenter confirms fixed display brightness, disabled auto-brightness/power-saving and variable refresh rate, unified browser fullscreen, closed high-load background processes, and an approximate 60 cm viewing distance. Refresh rate is re-measured after any display-mode change.

The browser records actual frame intervals in each trial's `requestAnimationFrame` loop; if the interval between adjacent callbacks exceeds $1.5\times$ the expected interval computed from the pre-trial measured refresh rate, a dropped-frame event is logged, along with estimated drop count, drop rate, mean/max frame intervals, observed refresh rate, basic subframe cycle rate, and full cycle rate with inversion frame. This recording reflects only browser presentation cadence and cannot substitute for panel scanning or photometric measurement.

Ethics and safety: The study is minimal-risk human-subjects research. The authors' institution does not maintain an ethics review committee for non-medical behavioral experiments of this type, so no institutional protocol number exists. Every session was administered in person by the experimenter under the safeguards described here. Before each session, the experimenter individually confirmed the participant's voluntary consent and photosensitivity-screening response in person. The web page then reiterated that the masked display involves rapid temporal modulation, that participants should stop immediately upon experiencing discomfort, eye fatigue, dizziness, nausea, or headache, and that they could withdraw unconditionally at any time. Proceeding recorded the confirmation timestamp and screening flag. Individuals who reported a history of photosensitive epilepsy or sensitivity to flicker were not eligible to continue. The implementation applies a conservative $f_r/n \geq 50$ Hz operating floor, but this engineering gate is not an ethical or clinical safety determination. At $n = 4$ on a 240 Hz panel the basic cycle rate is 60 Hz; the inversion frame reduces the full cycle frequency to $f_r/(n+1) = 48$ Hz, and the implementation separately logs warnings and actual timing. The formal study did not collect names, student identifiers, majors, ages, or gender information. Analysis and publication use only de-identified study records. Vision-correction status was collected to characterize the analyzed sample.

Task design: Each participant first completes one 12-second, unscored unmasked Canvas typing warm-up, views a 10-second, unscored readability-priority full-mask preview, and then completes an 8-second, unscored typing practice under the same readability-priority mask. This sequence separates initial stimulus and input adaptation from the four subsequent 20-second scored trials (within-subjects design). On every typing screen, including warm-up, practice, and scored trials, the stimulus Canvas is visible and its player has started before the participant clicks "Start." Clicking "Start" triggers a 5-second countdown while the input remains disabled; the timed typing interval and input availability begin only when the countdown ends. The same pre-start viewing procedure applies to both control and masked conditions.

The two conditions are each repeated twice. The experimenter's continuously assigned zero-based sequence number k selects ABBA or BAAB order to counterbalance practice and fatigue: (A) **original text** (control), displayed as unmasked Canvas with $n = 1$; (B) **masked text**, using the readability-priority configuration: $n = 4$, mask+noise, strong anti-OCR ($\alpha_s = 0.10$, $\alpha_g = 0.12$, 6 cycles) + weak inversion ($\alpha = 0.2$). The typing order index is $k \bmod 2$, and the subjective-rating Latin-square row is $\lfloor k/2 \rfloor \bmod 6$. Every consecutive 12 participants therefore cover all 2×6 joint assignments. Formal sessions enforce a uniqueness constraint on k, and the backend recomputes and verifies both order indices.

Both conditions are rendered by the same `renderSourceCanvas` path, with identical dark background, light text, canvas size, 23 px bold font, and color. The sole processing difference is the temporal protection pipeline, avoiding polarity and font-weight confounds. Here "cycles" means that the web implementation rotates through six distinct mask, noise, and stripe instances before repeating. This web-playback-specific setting shares α_s and α_g with the real-capture configuration in §IV and avoids repeatedly integrating one fixed overlay pattern.

Scoring no longer compares characters position-by-position but aligns the full transcription against the optimal target prefix using Levenshtein minimum string distance (MSD) [48]; untyped target suffix does not count as error. The alignment yields accuracy, characters per minute (CPM), and words per minute (WPM), along with edit distance, aligned prefix length, and scoring version. First-keystroke latency is defined from the end of the 5-second countdown, when the input is enabled and the scored timer starts, to the first input event that makes the transcription non-empty. Analysis first averages each participant's two trials per condition, then performs within-subject paired comparison.

Subjective ratings: After the typing trials, participants rate 6 conditions on the three 1–5 Likert items displayed by the interface: readability (1 = difficult to read, 5 = very clear), stability (stored in the `flicker` field; 1 = strong flicker, 5 = no perceptible flicker), and immediate visual comfort (`fatigue`; 1 = very uncomfortable, 5 = very comfortable). Higher values therefore indicate a better outcome on every item. Conditions include the unmasked original anchor, $n \in \{2, 3, 4\}$ mask+noise, $n = 4$ mask-only, and the same $n = 4$ readability-priority full configuration (mask+noise+strong anti-OCR+weak inversion) used in the typing task. The first three masking levels all achieve $\geq$60 Hz real temporal playback on a 240 Hz panel; the originally planned $n = 8$ would yield 30 Hz and trigger static fallback, so it is not included as a temporal ablation. The six conditions are presented in a 6th-order balanced Latin square; each condition must be viewed for at least 10 seconds before submission is enabled, with viewing start/end times and duration recorded. The unmasked anchor is also an attention/scale-interpretation aid but is not mechanically excluded by a "must rate 5" rule. This study does not include the high-suppression profile; assessment of its visible stripes and grain's acceptability is left to future work (see §VI).

TABLE 11. Typing Task: Within-Subject Comparison of the Control and Readability-Priority Masked Conditions ($N = 56$; Participant Means Over Two Trials per Condition; $'$ = Masked $-$ Control With Participant-Level Bootstrap 95% CI). Per Metric, a Paired t-Test Is Used When the Pre-Specified Shapiro Test Passes (Effect Size Cohen's d_z), Otherwise a Wilcoxon Signed-Rank Test (Effect Size Rank-Biserial r_{rb}); Test Statistics Are Reported in the Text

Metric	Control	Masked	Δ [95% CI]
Accuracy (%)	96.54	94.83	-1.71 [-3.39, -0.04]
WPM	20.44	19.30	-1.14 [-1.91, -0.39]
CPM	102.19	96.48	-5.71 [-9.56, -1.88]
Attempted chars	35.21	33.47	-1.74 [-2.88, -0.62]
First-key latency (ms)	742.64	833.61	$+90.96$ [-31.11, $+210.69$]

Sample size and analysis plan: A minimum of $N = 24$ participants was planned, a number within the range required for paired effect sizes of $d_z \approx 0.6$–0.8 at 80% power, and exactly covering two rounds of the 12-person joint order assignment. Pre-registered exclusion criteria are debug/demo sessions, incomplete four typing or six rating records, measured refresh rate below 200 Hz, any typing trial with fewer than 5 attempted characters, control mean accuracy below 50%, masked trials not in temporal mode, any trial with observed basic cycle rate below 50 Hz, or minimum-view straight-line ratings. The last criterion is defined as all six rating rows having a recorded viewing duration no greater than 11 seconds (a 1-second tolerance above the 10-second submission gate) and, within every row, identical scores on all three items. WPM, CPM, accuracy, attempted characters, and first-keystroke latency are first averaged within each participant by condition; speed metrics must be interpreted jointly with accuracy to avoid misinterpreting fast input under low accuracy as performance improvement. Paired differences are tested using paired t-tests when the pre-specified Shapiro test is passed, otherwise Wilcoxon signed-rank tests, with effect sizes and participant-level bootstrap 95% confidence intervals reported. Likert data are analyzed per dimension using Friedman tests with Holm-corrected post-hoc pairwise Wilcoxon tests. Analysis scripts generate de-identified participant means, JSON audit reports, and two LaTeX tables directly from the formal database `study_formal.db`.

Participants: 61 volunteers were recruited; 56 sessions met all pre-registered inclusion criteria and entered analysis (5 excluded: 3 for minimum-view straight-line ratings and 2 for control mean accuracy below 50%; the per-criterion audit accompanies the released analysis output). The 61 formal sessions filled each of the twelve joint order buckets five times, with one bucket receiving a sixth session. The formal study did not collect names, student identifiers, majors, ages, or gender information, so no age or gender distribution is reported; 46 of the 56 wore glasses, 1 wore contact lenses, and 9 reported no vision correction. All formal sessions ran on the 240 Hz laboratory display with a measured refresh rate of at least 200 Hz.

Results: Table 11 and Fig. 6 summarize the typing task. Relative to the control condition, the readability-priority

TABLE 12. Subjective Ratings for the Six Display Conditions ($N = 56$; 1–5 Likert, Higher Is Better; Mean $\pm$ SD). Stability Is the Inverse-Flicker Item and Comfort the Immediate Visual Comfort Item (§V-D); Friedman and Holm-Corrected Pairwise Wilcoxon Statistics Are Reported in the Text

Condition	Readability	Stability	Comfort
Unmasked anchor	4.80 ± 0.64	4.79 ± 0.68	4.75 ± 0.61
n=2 mask+noise	4.61 ± 0.68	4.77 ± 0.57	4.48 ± 0.87
n=3 mask+noise	4.21 ± 0.87	4.57 ± 0.76	4.11 ± 0.91
n=4 mask+noise	4.23 ± 0.91	4.38 ± 0.93	4.02 ± 1.05
n=4 mask-only	4.16 ± 0.87	4.45 ± 0.78	4.02 ± 0.96
Readability-priority full (n=4)	3.79 ± 1.17	2.98 ± 1.30	3.39 ± 1.25

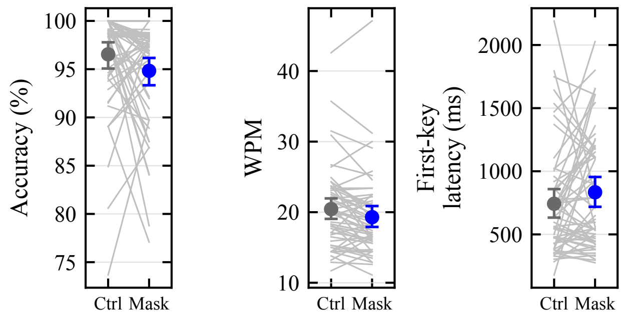

FIGURE 6. Within-subject typing performance under the control and readability-priority masked conditions: each gray line is one participant (mean of two trials per condition); markers show group means with participant-level bootstrap 95% confidence intervals. Panels show transcription accuracy, words per minute, and first-keystroke latency.

full configuration changed mean accuracy by -1.71 percentage points (95% CI $[-3.39, -0.04]$, Wilcoxon $W = 428$, $p = 0.043$, $r_{rb} = -0.33$), WPM by -1.14 (95% CI $[-1.91, -0.39]$, $t(55) = -2.93$, $p = 0.005$, $d_z = -0.39$), and first-keystroke latency by $+91.0$ ms (95% CI $[-31.1, +210.7]$, Wilcoxon $W = 641$, $p = 0.200$, $r_{rb} = 0.20$); CPM and attempted characters gave $p = 0.005$ ($d_z = -0.39$) and $p = 0.004$ ($d_z = -0.40$), respectively. Under Holm correction across the five typing metrics, WPM, CPM, and attempted characters remain significant whereas the accuracy difference ($p = 0.043$, rank 4 of 5) does not. Table 12 and Fig. 7 summarize the subjective ratings: Friedman tests yielded $\chi^2(5) = 72.2$ ($p = 3.5 \times 10^{-14}$) for readability, $\chi^2(5) = 123.2$ ($p = 6.5 \times 10^{-25}$) for stability, and $\chi^2(5) = 98.5$ ($p = 1.1 \times 10^{-19}$) for immediate visual comfort; Holm-corrected pairwise comparisons between the readability-priority full configuration and the unmasked anchor gave $p = 2.6 \times 10^{-4}$, $p = 2.2 \times 10^{-7}$, and $p = 1.8 \times 10^{-6}$ on the three items, respectively.

E. REAL-CAPTURE OBJECT DETECTION AND TRACKING

To probe whether the temporal degradation transfers beyond text, we report an exploratory detection/tracking stress test within the same camera link, noting that the evidence strength is not equivalent to the OCR main experiment. Content was displayed on the 240 Hz screen, captured by the S600 at 1.5 m / 0°, and processed by four detectors [49]–[52]. mAP, mAP50, and average recall (AR) are reported for detection. Tracking

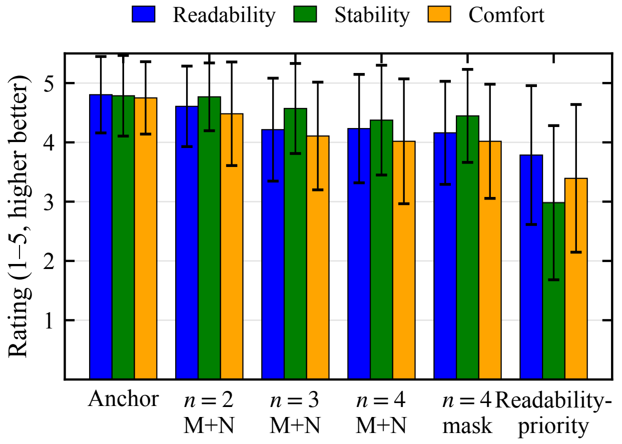

FIGURE 7. Mean 1–5 Likert ratings (error bars: SD) for readability, stability, and immediate visual comfort across the six rated display conditions, ordered from the unmasked anchor to the readability-priority full configuration. Higher is better on every item.

TABLE 13. Real-Capture COCO Detection (150 Images; `real_clean` = Unprotected Capture Baseline)

Detector	Condition	mAP	mAP50	AR
YOLO26x	`real_clean`	28.6%	43.9%	31.0%
	`real_short`	1.3%	2.4%	1.4%
	`real_video`	3.5%	5.3%	3.6%
RT-DETR-x	`real_clean`	30.1%	50.2%	34.8%
	`real_short`	3.5%	5.8%	4.1%
	`real_video`	4.6%	7.3%	5.5%
Faster R-CNN	`real_clean`	21.3%	39.2%	28.7%
	`real_short`	0.8%	1.6%	0.9%
	`real_video`	1.5%	3.0%	1.8%
RetinaNet	`real_clean`	22.6%	39.4%	27.0%
	`real_short`	0.9%	1.8%	1.0%
	`real_video`	1.9%	3.1%	2.0%

TABLE 14. Real-Capture MOT17-09-FRCNN Still-Frame Tracking (ByteTrack-Style Greedy Association, 450 Frames)

Detector	Condition	HOTA	IDF1
YOLO26x	`real_clean`	15.8%	12.1%
	`real_short`	8.7%	6.5%
	`real_video`	10.0%	8.7%
RT-DETR-x	`real_clean`	14.2%	10.4%
	`real_short`	8.1%	5.2%
	`real_video`	9.6%	6.8%
Faster R-CNN	`real_clean`	14.4%	9.6%
	`real_short`	6.8%	6.0%
	`real_video`	10.5%	7.1%
RetinaNet	`real_clean`	14.2%	9.8%
	`real_short`	5.1%	4.4%
	`real_video`	6.8%	5.9%

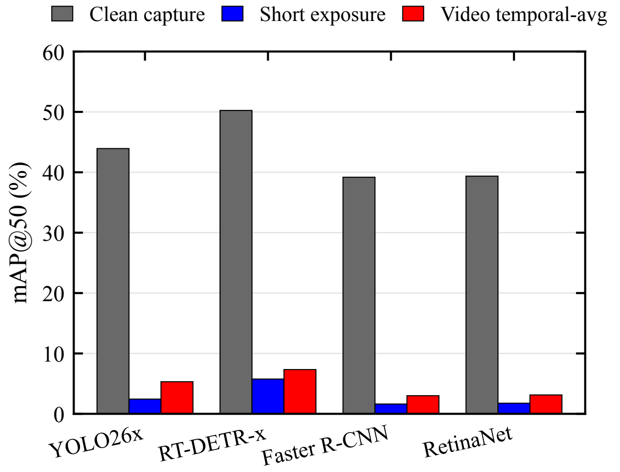

FIGURE 8. Real-capture mAP50 for four detectors across the unprotected capture, short-exposure protected capture, and 150-frame protected mean. Conditions differ in protection, exposure, resolution, and frame reduction.

uses higher-order tracking accuracy (HOTA), identification F1 (IDF1), multiple-object tracking accuracy (MOTA), and multiple-object tracking precision (MOTP). The protected condition uses the $n = 4$ mask+noise profile without overlays or inversion. The `real_clean` reference is an unprotected nominal 31.25-ms capture. The `real_short` image is the maximum-grayscale-contrast frame from a three-frame protected burst at the nominal 3.91-ms setting, and `real_video` is the mean of 150 protected frames acquired nominally at 60 fps. Tracking uses 450 frames from MOT17-09-FRCNN via still-display-then-capture and an in-project greedy association implementation inspired by ByteTrack [53].

Under short exposure, all four detectors' mAP50 drops from 39.2–50.2% to 1.6–5.8% (Table 13, Fig. 8), representing relative decreases exceeding 85%, while the `real_video` condition yields 3.0–7.3%, also below the `real_clean` baselines. Because protection, exposure, resolution, and frame reduction differ across conditions, these values form a cross-task stress test rather than an isolated protection effect. In still-frame tracking (Table 14), HOTA decreases from 14.2–15.8% to 5.1–8.7% and IDF1 from 9.6–12.1% to 4.4–6.5%.

F. SOFTWARE SIMULATION EXPERIMENTS

This section uses software simulation to address two questions: (1) what is the machine readability of one protected subframe obtained after the application renderer, and (2) what is recovery under zero camera noise and perfect alignment? By characterizing the post-render single-subframe capture point, the experiment defines a boundary that malware intercepting the original frame or accumulating a complete cycle would bypass.

1) Synthetic Single-Subframe OCR

Using Tesseract [42], EasyOCR [41], and Surya [40] on 120 synthetic samples, character recovery drops from 94.0–97.0% to 0–2.6% across three engines (Table 15). Stratified

TABLE 15. Synthetic Single-Subframe OCR: Three-Engine Character Recovery (120 Samples, $n = 4$, $\varepsilon = 8/255$)

OCR engine	Original frame	Single subframe	Reduction
Tesseract	94.0%	0.0%	94.0%
EasyOCR	94.1%	0.0%	94.1%
Surya	97.0%	2.6%	94.4%

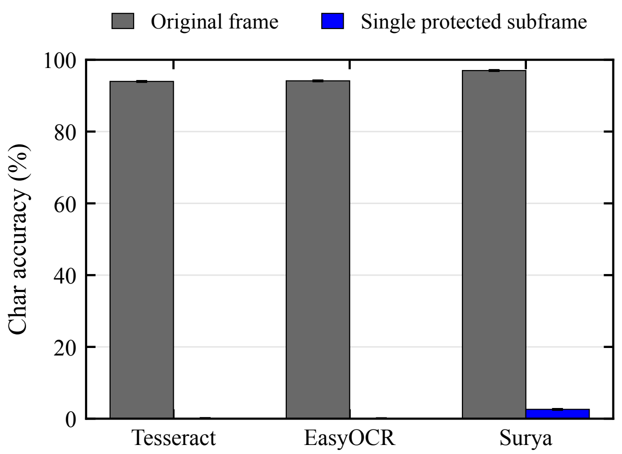

FIGURE 9. Character recovery for three OCR engines on original frames and single protected subframes; error bars are 95% bootstrap intervals resampled by corpus sample.

breakdowns further characterize recovery across the evaluated content categories.

Fig. 9 visualizes these results; single-subframe recovery does not exceed 2.6% for any engine.

2) Theoretical Security Boundary

When an attacker accumulates registered frames covering a full cycle, complementary subframes can be re-summed in the ideal linear model of (1), with the archived per-sample strongest digital attack recovering 95.2% and pure full-cycle temporal averaging recovering 94.3% (Table 16)—values that empirically mark the reconstruction boundary of the evaluated linear decomposition. The high-suppression profile yields 47.9% pooled recovery under the 150-frame mean and 53.9% after excluding the underexposed position. Character recovery also depends on content and the physical imaging path.

3) Detection and Tracking Simulation

Simulations were run on COCO val2017 [54] and MOT17 [55] with YOLO26x [49], RT-DETR-x [50], Faster R-CNN [51], and RetinaNet [52]: the COCO simulation used 8 images as an implementation pipeline diagnostic (Table 17), while the MOT simulation covers 7 sequences totaling 5,316 frames and uses the same in-project ByteTrack-inspired greedy association method (Table 18).

On the 8 COCO images, single-subframe mAP50 across the four detectors is 0–6.0%. Across the 5,316 MOT frames,

TABLE 16. Archived Digital Reconstruction Boundary (Tesseract)

Setting	N	Char.	Exact	SSIM	ΔE_{00}
Base: cycle mean	120	94.3%	85.8%	–	–
Base: per-sample strongest	120	95.2%	91.7%	–	–
Readability: cycle mean	16	87.9%	50.0%	0.948	1.46
High: cycle mean	16	56.6%	6.3%	0.764	4.87
High: inversion slot	16	93.0%	81.2%	–	–

The 120-sample base attack and 16-sample profile ablation are separate archived digital evaluations. "Per-sample strongest" selects the best result among seven prespecified reconstructions (temporal average cycle, blue channel max, differential blue, differential luma, global shutter slot 0, phase search max, and phase search mean) for each sample. "Inversion slot" extracts only the $\alpha(\mathbf{1} - \mathbf{I})$ frame and applies contrast stretching; its 93.0% recovery identifies slot-level composition as a critical design variable. Profile SSIM and ΔE_{00} describe full-cycle reconstruction quality, not physical or user-perceived quality.

TABLE 17. COCO val2017 Simulated Detection Pipeline Diagnostics (4 Detectors, 8 Images)

Detector	Condition	mAP	mAP50	AR
YOLO26x	clean	48.1%	58.3%	51.0%
	single subframe	0.0%	0.0%	0.0%
	temporal avg.	14.3%	18.3%	14.3%
RT-DETR-x	clean	47.1%	58.7%	53.4%
	single subframe	5.2%	6.0%	5.2%
	temporal avg.	16.8%	23.7%	17.9%
Faster R-CNN	clean	38.2%	52.7%	43.6%
	single subframe	3.0%	4.4%	3.0%
	temporal avg.	11.1%	16.5%	11.1%
RetinaNet	clean	33.2%	47.1%	35.3%
	single subframe	3.3%	4.2%	3.3%
	temporal avg.	9.6%	13.2%	9.6%

TABLE 18. MOT17 Simulated Tracking (ByteTrack-Style Greedy Association; 5,316 Frames)

Detector	Condition	MOTA	MOTP	IDF1
YOLO26x	clean	31.1%	81.7%	27.1%
	single subframe	0.1%	74.8%	0.1%
	temporal avg.	9.1%	78.1%	12.0%
RT-DETR-x	clean	24.1%	79.8%	38.2%
	single subframe	−2.3%	73.2%	5.4%
	temporal avg.	−11.4%	75.0%	14.7%
Faster R-CNN	clean	3.8%	77.9%	35.6%
	single subframe	−2.5%	71.2%	4.5%
	temporal avg.	−2.7%	73.2%	14.0%
RetinaNet	clean	−5.4%	77.1%	30.5%
	single subframe	0.4%	70.6%	2.7%
	temporal avg.	−2.9%	73.8%	11.9%

single-subframe IDF1 is 0.1–5.4%, while MOTA includes negative and nonmonotonic values. Negative MOTA occurs when the false-positive, false-negative, and identity-error penalties exceed the number of ground-truth objects; it is not a negative detection probability. Detector architecture, confidence thresholds, proposal density, and the greedy associator jointly shape the observed differences. Fig. 10 summarizes the residual capability of a single protected subframe relative to the clean baseline across the evaluated OCR, detection, and tracking pipelines.

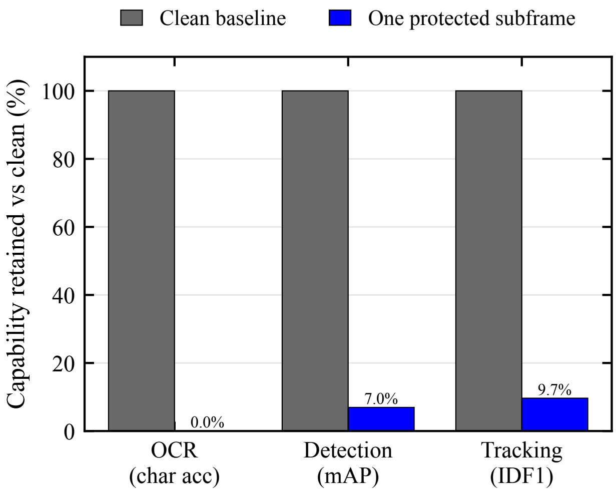

FIGURE 10. **Residual capability of one protected subframe relative to the clean baseline (clean = 100%) across the evaluated OCR, detection, and tracking pipelines. The metrics are normalized within task and are not directly commensurate across tasks.**

VI. DISCUSSION

A. PRINCIPAL FINDINGS AND RESEARCH VALUE

At the common S600 setting, readability-priority recovery is 17.8%, compared with 94.5% for unprotected content. High-suppression further lowers recovery to 5.6%. The paired content-cluster gaps remain 76.7 and 88.9 percentage points across 288 matched units per profile. A reproducible preprocessing grid raises protected recovery, yet large paired gaps remain. Detection, tracking, VLM, integration, and simulation experiments then map how attacker capability changes these outcomes.

Relative to prior temporal-display studies, the 10,575-capture physical archive is new, as is the link from average OCR recovery to explicit-field leakage and upper-tail risk and the evaluation of attacker escalation across preprocessing, sustained capture, and modern recognition models.

B. INTEGRATION AND ATTACKER ESCALATION

Temporal masking exploits the difference between human temporal integration and short camera exposure; in the ideal linear model, registered complementary subframes can reconstruct the original because $\sum \mathbf{I}_k = \mathbf{I}$, and the digital attack evaluation confirms this boundary, reaching 95.2% per-sample strongest recovery and 94.3% under pure full-cycle averaging.

Physical temporal averaging also increases recovery without requiring the mask seed or phase: the 150-frame mean reaches 71.1% for readability-priority and 47.9% for high-suppression, rising to 79.7% and 53.9% at the common settings, so that sustained capture converts temporal fragments into a usable recognition signal.

C. PROFILE TRADE-OFFS AND PRACTICAL IMPLICATIONS

The two principal profiles occupy distinct operating points: readability-priority preserves a digitally reconstructed SSIM of 0.948 but retains 17.7% explicit-field recovery, whereas high-suppression lowers several recovery measures while reducing digital SSIM to 0.764 and increasing ΔE_{00} from 1.46 to 4.87, making the privacy–readability trade-off between these profiles explicit and tunable.

Because the high-suppression inversion slot reaches 93.0% recovery after contrast stretching, slot-level composition remains a weak point in the current design, and a practical system should therefore combine temporal masking with field-level policies and independent access controls for high-value content.

D. VLM-AWARE IMPLICATIONS

Modern VLMs recover far more text than conventional OCR from the same protected photographs, with recovery strongest for large-font credentials, code, digits, and short sentences while dense CET6 pages remain more strongly suppressed. The implication is that temporal profiles should adapt to glyph scale, semantic predictability, and field sensitivity.

E. ETHICS STATEMENT

This research is defensive in nature, aiming to protect screen content from unauthorized photography. The attack analysis (§V-F2) is intended for honest assessment of security boundaries, not as an attack guide. The user study involved human subjects in minimal-risk behavioral research. The authors' institution does not maintain an ethics review committee for non-medical behavioral experiments of this type, so no institutional determination was available. Before each session, the experimenter individually confirmed voluntary consent and photosensitivity-screening eligibility in person; the web interface then recorded the confirmation state and timestamp. The protocol also included in-person supervision and voluntary withdrawal as described in §V-D. Analysis and publication use only de-identified data.

F. LIMITATIONS AND FUTURE WORK

The physical evidence comes from one S600 UVC webcam, and profile order was fixed rather than randomized. Exposure values are control-derived labels rather than photometric shutter measurements. The archive also lacks decoded auto-exposure state, gain, white-balance, focus, ambient illuminance, display–camera phase, and physical panel luminance. These factors can interact with time, geometry, and profile order.

The current PoC operates at the application layer with fixed panel output. It lacks photometric slot-timing measurements, luminance-matched static controls, and matched noise-off versions of the enhanced profiles. Temporal sparsity, duty-cycle luminance, overlays, panel response, camera exposure, and image signal processor (ISP) processing therefore remain coupled. Rolling-shutter row mixing and panel transitions are candidate explanations for the simulation-to-real gap.

The content-cluster intervals hold the tested geometries fixed, and the 12-item corpus is too small for script-level inference. Detection uses 150 COCO images at one position, while tracking uses 450 still-frame captures from one MOT17 sequence. The VLM probes use one on-axis profile, three API endpoints, and raw JPEG inputs. Commercial OCR, learned restoration, and unconstrained adaptive preprocessing may recover additional text. The 150-frame experiment also leaves the duration–recovery curve unresolved.

Future work should randomize profile order, lock camera parameters, record batch time, and measure physical slot timing and luminance. Matched static controls can separate temporal and luminance effects. Replication across cameras and displays can test transfer, while continuous-video attacks can map recovery against recording duration. VLM-aware profiles can explore font-size-adaptive granularity, glyph decoys, and field-level rendering. Extending the usability study to longer sessions, other panels, and the high-suppression profile would map comfort beyond the present short-duration evaluation.

VII. CONCLUSION

We evaluated temporal pixel masking on one UVC camera link. Across 8 common-setting geometries, the same 288 matched units per profile (12 content items $\times$ 8 geometries $\times$ 3 repeats) yield 94.5% mean recovery unprotected, 17.8% for readability-priority, and 5.6% for high-suppression. Paired content-cluster contrasts are 76.7 percentage points ([74.3, 79.2]) and 88.9 points ([85.5, 91.9]). Readability-priority retains 17.7% matched field-micro recovery and 95.0% P99 character recovery, while high-suppression meets its descriptive $< 10\%$ target.

Attack-oriented evaluation recovers substantially more information: a 150-frame temporal mean reaches 71.1% character recovery for readability-priority and 47.9% for high-suppression, rising to 79.7% and 53.9% at the common settings; in the full 9-geometry sensitivity pool, long-exposure recovery is 60.9% for readability-priority versus 47.3% unprotected. On-axis VLM probes also recover considerable large-font content, while the three-engine preprocessing oracle reaches 40.2% for readability-priority and 13.7% for high-suppression.

In a 56-participant user study, readability-priority masking retained 94.83% typing accuracy while reducing speed by 1.14 WPM; subjective readability, stability, and comfort averaged 3.79, 2.98, and 3.39 out of 5, confirming measurable usability costs concentrated in perceived stability for the readability-priority operating point. Detection and tracking stress tests show the same directional OCR-reduction pattern across four detectors. Future work should extend capture to additional cameras and displays, map VLM-aware adaptive profiles, and evaluate comfort over longer sessions.

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